



When I ask people to name three recently implemented technologies that most impact our world today, they usually propose the computer, the Internet, and the laser. All three were unplanned, unpredicted, and unappreciated upon their discovery, and remained unappreciated well after their initial use. They were consequential. They were Black Swans. Of course, we have this retrospective illusion of their partaking in some master plan. You can create your own lists with similar results, whether you use political events, wars, or intellectual epidemics.

You would expect our record of prediction to be horrible: the world is far, far more complicated than we think, which is not a problem, except when most of us don't know it. We tend to "tunnel" while looking into the future, making it business as usual, Black Swan-free, when in fact there is nothing usual about the future. It is not a Platonic category!

We have seen how good we are at narrating backward, at inventing stories that convince us that we understand the past. For many people, knowledge has the remarkable power of producing confidence instead of measurable aptitude. Another problem: the focus on the (inconsequential) regular, the Platonification that makes the forecasting "inside the box."

I find it scandalous that in spite of the empirical record we continue to project into the future as if we were good at it, using tools and methods that exclude rare events. Prediction is firmly institutionalized in our world. We are suckers for those who help us navigate uncertainty, whether the fortune-teller or the "well-published" (dull) academics or civil servants using phony mathematics.

From Yogi Berra to Henri Poincaré

The great baseball coach Yogi Berra has a saying, "It is tough to make predictions, especially about the future." While he did not produce the